Sensing Domestic Activity with Microphones

Sarah Clark
160505218
Dr Eliane Bodanese
MSc Artificial Intelligence

Abstract— **Being able to identify domestic activity can have many applications such as assistive technology for disabled people, remote monitoring for elderly people, and security applications. Sensing technology for home environments has been investigated using smart phones, cameras, and other sensors, however many of these are quite intrusive and often people are not comfortable being filmed at home. Therefore, an audio system for sound classification could be much more subtle for everyday use and more widely accepted in the home. This project looks at methods of audio feature extraction and classification using a Raspberry Pi computer and a microphone to classify domestic activity. Several different classification methods were investigated for this task: a K-Nearest Neighbours (KNN) classifier, a Support Vector Machine (SVM) and several different Convolutional Neural Network (CNN) models. The KNN and SVM models achieved the best accuracy on the test data, but a Resnet CNN model proved to be the most effective for this task when applied to real life classification, likely because the CNN models had been trained using data augmentation techniques to make them more resilient to noise.**

I. INTRODUCTION

Over recent years, there has been a rising interest in smart home environments in which humans are able to control and monitor their homes for reasons related to security, comfort, and the environment. These smart devices use sensors to monitor their environment and receive useful information which can then be used and applied to enhance daily living. Audio data is an example of this useful information that can be interpreted, for example used by voice assistants such as Google Home and Amazon Echo which focus on speech.

Audio data, however, can also be used to identify other domestic activities performed by humans in the home, with several different applications. These include assistive technology for people with hearing impairments, such as alerting them to warnings/alarms, a doorbell, or a ringing phone; security applications; and most significantly as remote monitoring systems for elderly or disabled people. An increase in the average life expectancy during the past few decades (Suzman & Beard, 2015) which is expected to continue over the next century (United Nations, 2015) means that there are more and more elderly people living alone. Remote monitoring can be used to collect information to track behaviour patterns of individuals as they go about their daily lives to ensure that they are completing routine activities and looking after themselves.

Audio data for monitoring systems have several advantages over video systems. Firstly, audio data is less computationally intensive than video data as it is one-dimensional (time) compared to three dimensions (width x height x time). Microphones are also generally less expensive and smaller than video cameras meaning they are more suited to domestic systems. Finally, audio surveillance is less invasive than video surveillance in terms of a user's privacy.

This project looks at recording and identifying domestic sounds using microphones. In general, the task of sound recognition, also known as Acoustic Event Recognition (AER), is performed by extracting features from the audio and training classifiers to recognise features of audio belonging to each desired class. In real environments, recognising audio patterns is made difficult as sound events can be unstructured, unpredictable, and diverse (Chu, et al., 2006). This can cause more challenges compared to audio tasks such as speech and music signal processing, and it is difficult to use traditional sound characteristics to represent these signals. Therefore, many studies have looked at feature extraction methods for sound events, with time-frequency representations proving effective in many cases (Sharan & Moir, 2019; Ghoraani & Krishnan, 2011).

Similarly, many studies have looked at different classification methods for AER such as Gaussian Markov Models (GMMs), Hidden Markov Models (HMMs), K-Nearest Neighbours classification, Support Vector Machines (SVMs) and Neural Networks (NNs) (Stowell, et al., 2015). In recent years, Deep Neural Networks (DNNs) have proved more popular (Mesaros, et al., 2018), in particular using Convolutional Neural Network (CNN) models to classify different sounds.

This project looks at developing an AER system for a small number of domestic sound events considering the best approaches here for audio feature extraction and classification using live recordings. The paper is organised as follows: Section II presents an overview of relevant audio feature extraction and classification literature. Section III describes the methods implemented to create the final system. Section IV looks at the results from the implemented system, and the final sections analyse the results and look at what conclusions can be found.

II. Related Work

A. Feature Extraction

Audio feature extraction techniques affect the performance of a system. Many different audio extraction methods have been investigated.

Audio signals are commonly transformed into an image representation, such as a time-frequency representation which plots the spectrum of frequencies against time. Classifiers can then learn patterns in the signal in the same way as image classification methods.

Most audio classification systems use time-frequency representations based on the Short-Term Fourier Transform (STFT) such as Mel-Frequency Cepstral Coefficients (MFCCs) (Logan, 2000; Boddapati, et al., 2017), chromograms (Su, et al., 2019), and most commonly log-scaled Mel-spectrograms which are found to be successful as they provide a good representation of the spectral properties of a signal (Sakashita & Aono, 2018; Nguyen & Pernkopf, 2018).

Both Han et al. (2017) and Sakashita & Aono use Mel-spectrograms as audio features, also taking Mel-spectrograms of Harmonic-percussive source separation (HPSS) applied to a mono input and using all spectrograms as model input channels. The idea behind using HPSS features is that for some classes, the harmonic and percussive content of the audio data may have different characteristics. These additional audio features were found to help improve classification accuracy.

B. K-Nearest Neighbours Classification

Audio classification tasks have previously been performed successfully with K-nearest neighbour classifiers (Mahajan, et al., 2009). This is a supervised machine learning approach which assigns a class to unseen data based on the most common class of its k-nearest neighbours, when looking at the audio features extracted. This algorithm makes the assumption that data with similar features will belong to the same class, and will therefore exist in close proximity in the feature vector field.

C. Support Vector Machine Classification

Support Vector Machines are another classification method which has worked well with audio classification tasks before (Lu, et al., 2003). SVMs created hyperplane decision boundaries to separate classes. The objective is to find a decision boundary that has the maximum distance between data points belonging to both classes. For audio data this means that feature vectors can be classified in multidimensional space by drawing hyperplane boundaries.

D. CNN Network Architecture

Deep Neural Networks (DNN) have been studied for audio classification applications, with convolutional Neural Network (CNN) architectures proving effective in audio classification (Hershey, et al., 2017; Kong, et al., 2020). CNNs are effective at extracting local features and patterns, and can help capture spatial information which is more important in audio applications than time series data (Sakashita & Aono, 2018).

Deep convolutional neural networks are commonly used for image classification problems as they have been found to produce high accuracies (Krizhevsky, et al., 2012; Zeiler & Fergus, 2014; Sermanet, et al., 2014). Many studies have compared models and found that network depth is an important factor for learning features (Simonyan & Zisserman, 2015; Szegedy, et al., 2015), and therefore improving classification results. However, the question of whether adding more layers will lead to improved models, has led to further problems, such as overfitting, the fact that deeper networks require more parameters, and therefore have more memory and computational requirements, in particular during training, and the problem of vanishing gradients (Bengio, et al., 1994; Glorot & Bengio, 2010). The vanishing gradient problem is caused by saturating neurons, e.g. activations that tend toward 1 at high values, and tend toward 0 at low values, such as sigmoid and tanh. Once saturated, the model struggles to adapt weights further and improve learning. This is a problem particularly in deep models which have many layers as weights are updated by backpropagating error through the network, and the amount of error dramatically decreases at each layer due to the derivative of the activation function. The ReLU activation is common in deep learning as it is better at dealing with the vanishing gradient problem since the gradient of the ReLU function is always either 0 or 1 and can never saturate, and therefore the gradients can't vanish.

Image classification networks, such as VGGNet, ResNet and Inception have been found to perform well on audio classification tasks (Hershey, et al., 2017; Guzhov, et al., 2021). The VGGNet architecture proposed by Simonyan et al. (2015) looked at increasing depth by replacing large kernel-sized filters used in previous architectures (11x11, 5x5) with multiple fixed size kernels of size 3x3. This allowed the depth of the network to be increased so that more complex features could be learned, but less parameters were used so the cost was reduced. Still, VGG has a large computational requirement due to the large width of convolutional layers, and this also makes it slow to train.

He et al. (2016) proposed ResNet which uses short-cut connections to overcome the vanishing gradient problem and create an extremely deep network. ResNet uses Residual blocks, and learns the identity mappings between these, and this allows networks to be training deeper without exploding gradients. This also addresses the degradation problem in which accuracy gets saturated and then begins to degrade.

A new trend in deep CNN architecture was the shift to modular uniform layer design to make it easier for CNNs to be built for different tasks easily. The GoogLeNet, also known as Inception, architecture (Szegedy, et al., 2015) uses an inception module that approximates a sparse CNN with a normal dense construction. This is based on the knowledge that, as most of the activations in a deep network either have a value of zero, or redundant due to correlations between them, not all of the output channels will have a connection with the input channels. Therefore, not all neurons are needed, and so the inception module helps to reduce the kernel size and therefore limit computational requirements. The Inception module also uses wider kernels, implementing multiple kernels of different sizes within the same layer to recognise features of different sizes.

Several papers have investigated variations on popular image classification DNNs for audio applications, such as Han et al. (2017) who base their CNN architecture on VGGNet, and Guzhov et al. (2021) who propose a model based on ResNet-50. Both show that these networks designed for image classification can perform well on audio data as well, although Han et al. observe that due to a smaller dataset of audio data, using residual connections to increase the number of layers in the deep network is not particularly effective. Nonetheless, Hershey et al. (2017), who investigate image classification networks on several audio tasks, show good results for many image classification networks, with the best results achieved by Inception and ResNet which are able to capture common structures for audio as well as images.

One important consideration is that improvements to networks to increase accuracy do not necessarily take network efficiency, such as speed and computation, into consideration. For real world applications, recognition tasks generally need to be carried out on devices with limited computational power and so the complicated DNN models used previously are not suited to most mobile and embedded systems. MobileNet proposed by Howard et al. (2017) is a network that aims to reduce the number of model parameters and multiply-add operations by using depth wise separable convolutions. It uses residual connections like ResNet, but is much more computationally efficient. Kong et al. (2020) adapted MobileNetV1 (Howard, et al., 2017) and MobileNetV2 (Sandler, et al., 2018) for audio applications and find that both models achieve good performance and are much more computationally efficient than other networks used. They find MobileNetV1 slightly more accurate than MobileNetV2, but MobileNetV2 to be the most efficient of the models tested.

E. *Data Augmentation*

In order to learn a non-linear function that generalizes well on unseen data, deep neural networks need a large quantity of training data. It has been shown that a larger labelled dataset can improve accuracy for deep networks (Mun, et al., 2017), however collecting a large, labelled dataset can be challenging. A method to combat this is the use of data augmentation, i.e. increasing the amount of training data through various means such as transforming data and creating new synthetic data. Data augmentation can help to increase the model's robustness to unknown data and reduce overfitting. Previous studies have used augmentation techniques such as random combinations of time stretching (changing the speed of the audio sample without changing the pitch), pitch shifting (changing the pitch of the audio without changing the speed), time shifting and also mixing the sample with background noise (Piczak, 2015; Parascandolo, et al., 2016; Salamon & Bello, 2017). Salamon and Bello report pitch shifting to be the most effective, whereas Piczak finds that none of the data augmentation techniques make a noticeable improvement to training.

Mun et al. (2017) used Generative Adversarial Networks (GANs) to perform data augmentation by creating new synthetic data samples. A GAN model, first proposed by Goodfellow et al. (2014) for image generation consists of two networks: a generator network and a discriminator network. The generator takes noise, z, as an input and outputs a generated data sample, G(z). The discriminator takes in a data sample and outputs a score based on whether the input is real or fake (i.e. output from the generator). The aim of the discriminator is to identify which features belong to a real data sample, and which features belong to a fake data sample. The aim of the generator is to learn how to avoid the features that the discriminator has learned to recognise. Muns et al. applied this approach to audio data and find that it greatly improved the classification accuracy, achieving Top Rank in DCASE 2017 Task.

Another important data augmentation method used by almost all entries in Task 1 of the DCASE challenge for all years since 2019 (DCASE Challenge, 2019; 2020; 2021) is 'mix-up', which mixes a pair of two randomly selected training samples to create a new virtual training sample, according to a beta distribution (Zhang, et al., 2017). The new mixed samples are used for training, and the training target is the mixing ratio. Zhang et al. (2018) use mix-up for environmental Sound Classification (ESC) with a deep neural network architecture and find that it improved class accuracy and generally reduced variance within classes.

III. Methodology

A. Audio Data

In order to build a dataset of domestic sounds covering these classes, audio data was compiled from a number of different datasets. From the ESC-50 dataset (Piczak, 2015), audio data was used belonging to the classes 'toilet flushing', 'vacuum cleaning', 'keyboard typing' and 'coughing'. Audio samples belonging to the classes 'toilet flushing', 'keyboard typing' and 'coughing' were taken from the FSD-50K (E. Fonseca, 2020), 'vacuum cleaner' samples were taken from the DESED dataset (Turpault, et al., 2019), and samples belonging to the 'neutral' class were taken from the Chime-home dataset (Foster, et al., 2015). From Google's Audioset (Gemmeke, et al., 2017) audio samples were used for the classes 'door knocking', 'toilet flushing', 'vacuum cleaning', 'keyboard typing', 'coughing' and 'neutral'. All other audio data was downloaded from the Freesound database of Creative Commons Licensed Sounds (Font, et al., 2013) or recorded for the project. Synthetic data was also created using 'Scaper' (Salamon, et al., 2017).

In total, there were 19,065 audio data samples (~318 minutes) split according to the classes shown in Table 1. The training/testing split was 80/20. In order to address class imbalance, a Weighted Random Sampler was utilised during training.

TABLE I. AUDIO DATA SAMPLES

Domestic Sound Class	Number of Data Samples
Door Knocking	2200
Shower Running	2981
Toilet Flushing	2381
Vacuum Cleaning	3363
Keyboard Typing	2623
Coughing	2627
Neutral	2890

B. Data Pre-processing

All audio samples were pre-processed using functions from the Librosa library. First, data was resampled at 44.1 kHz, then it was converted to mono, as not all audio samples had multiple channels, by averaging samples across channels. Next audio data was resized to one second, with any sample that was too long being trimmed. Finally, the model inputs were normalised.

B. Audio Feature Extraction

After the data samples had all be standardised to the same size, audio features were extracted. For the KNN and SVM models, the audio data was converted to a log-scaled Mel-spectrogram representation with 64 Mel bands and an FFT window size of 1024. HPSS was applied to these Mel-spectrograms using the Librosa library, to produce two further spectrograms. Librosa functions were also used to produce further audio features:

- The zero crossing rate: Calculated as the number of times the signal crosses from positive per negative per frame, with frame length of 1024. Librosa returns this as a Numpy array.
- The spectral centroid: Librosa returns a Numpy array containing the mean of each frame of a normalised magnitude spectrogram.
- Mel-Frequency Cepstral Coefficients (MFCCs)

For the input to the CNN models, only Mel-spectrogram and HPSS spectrogram features were used. Data augmentation was also applied such as pitch shifting by randomly shifting audio data up or down pitches in the range ±3. Mix-up was also applied to the input spectrograms.

C. Hardware

In order to produce the AER system to recognize domestic sounds multiple models were trained and tested. The CNN models were trained using a NVIDIA Tesla K80 GPU. Due to GPU constraints, models were only able to be trained for 30 epochs. After training the models were saved to be used on a Raspberry Pi 4 Model B computer. A ReSpeaker 4-Mic Array microphone was attached to the Raspberry Pi in order to record sounds live which could then be classified using the trained models. For this project one second samples of audio were again used at the live recording stage in order to test the performance of the system.

D. Classification Model

Several different classification models were investigated for this task: a SVM classifier, a KNN classifier, a 14-layer CNN proposed by Kong et al. (2020), based on the VGG network used in image classification, a ResNet model adapted for audio classification (Kong, et al., 2020), and a MobileNetV2 model adapted for audio classification (Kong, et al., 2020).

The CNN models were trained by optimizing the cross-entropy loss and the Adam optimizer with a learning rate of 0.001 was used. The batch size was set to 30 and the data was shuffled between epochs. The dataset was split 80% training set and 20% validation set.

IV. RESULTS

A. Validation Dataset

The results form testing the models on the validation dataset can be seen in Table II. The KNN and SVM models were the fastest to produce as they did not require training in the same way as the CNN models. They also produced the highest accuracy on the validation dataset. The KNN model was found to be slightly more accurate than the SVM model, however it was also a larger model that was marginally slower to make a prediction.

The CNN models did not perform as well on the validation dataset as the KNN and SVM models. In the cases of the CNN-14 and MobileNet models, the testing set accuracy was noticeably higher than the validation set accuracies suggesting that these models had suffered from overfitting during training, and this was why they performed poorly on the unseen data. However, when looking at the class confusion matrixes plotted from these models (Figures III and V) there does seem to be some learning with sound classes being recognised generally in two groups which seem to be 'percussive' sounds and 'continuous' sounds. The ResNet proved more successful and performed almost as well on the unseen data as it did on the training dataset.

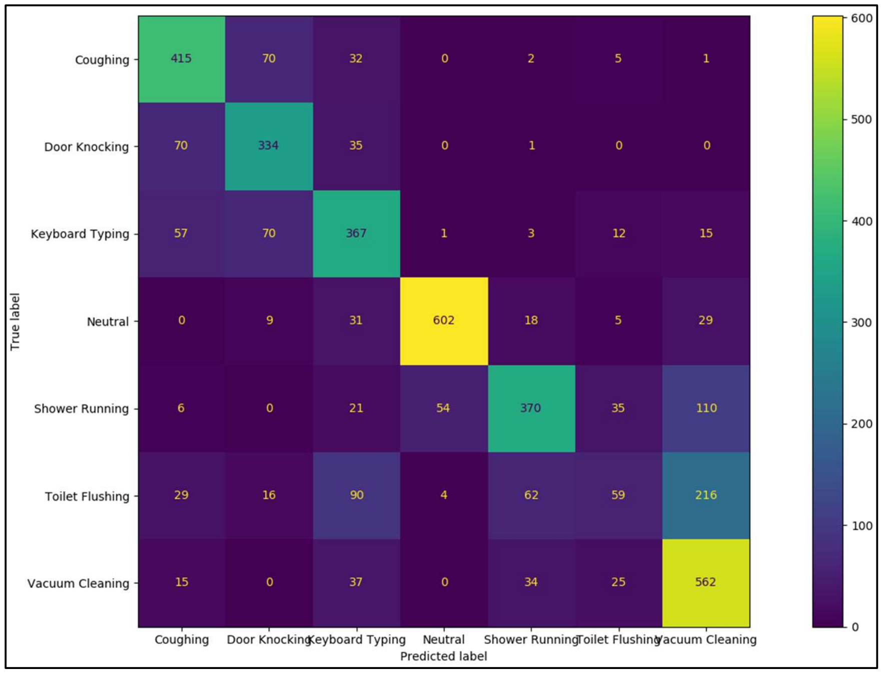

Figure I – Confusion matrix for SVM model tested on validation dataset.

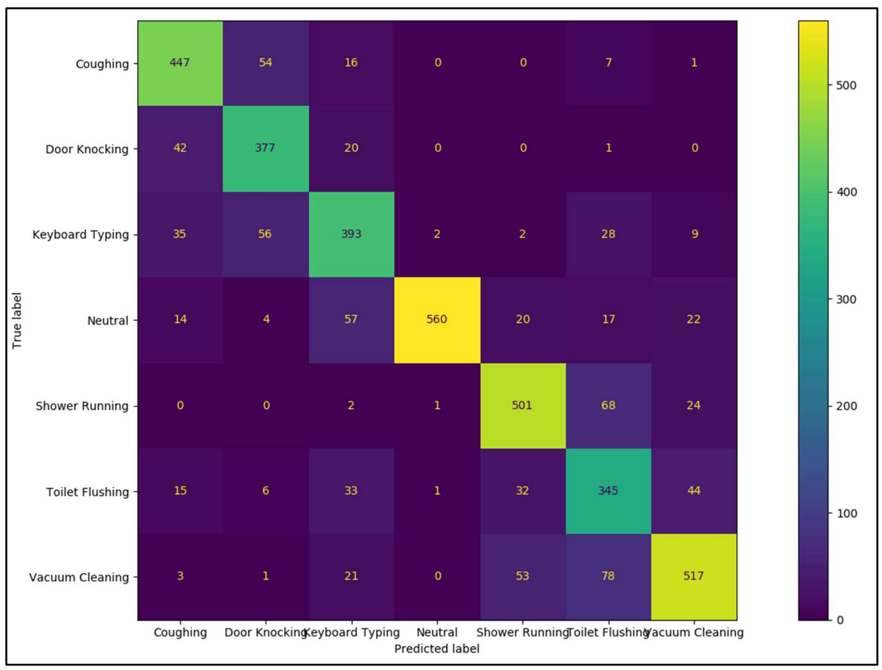

Figure II – Confusion matrix for KNN model tested on validation dataset.

In terms of the size of the models, the CNN-14 model was the largest and proved very slow to load so that it

TABLE II. MODEL CLASSIFICATION RESULTS

Classification Model	Test Set Accuracy	Validation Set Accuracy	Model Size (KB)	Prediction Time (s)
KNN	N/A	79.9%	1,095	0.14
SVM	N/A	68.9%	750	0.03
CNN-14	55.9	35.3%	622,850	2.59
ResNet	56.6%	51.7	200,260	0.87
MobileNet	70.8%	33.7%	17,850	1.54

could be used. As expected the MobileNet model was the most efficient, being much smaller in size than the others.

B. *Raspberry Pi System*

All of the models were tested on the Raspberry Pi system with live audio data. To provide a fair comparison, each audio class was tested on each model 10 times and the accuracy of their predictions were recorded. This can be seen in Table III.

Interestingly, at this point the KNN and SVM models were much less successful. The SVM model in particular continually only output predictions in one class, and the KNN model performed well on the ‘continuous’ sounds, but found it difficult to identify ‘percussive’ sounds.

As the CNN-14 and MobileNet models seemed to have overfit, predictably they did not provide good classification results. The MobileNet model predicted ‘Toilet Flushing’ nearly every time, although occasionally recognised some other ‘continuous’ sounds. The CNN-14 model was a bit more successful with some success in some classes, but still gave wrong predictions in most cases.

However, the ResNet model provided quite high accuracy when presented with an audio class. In particular this model performed well on the more percussive sounds, i.e. ‘Door Knocking, generally 100% correct on the live audio

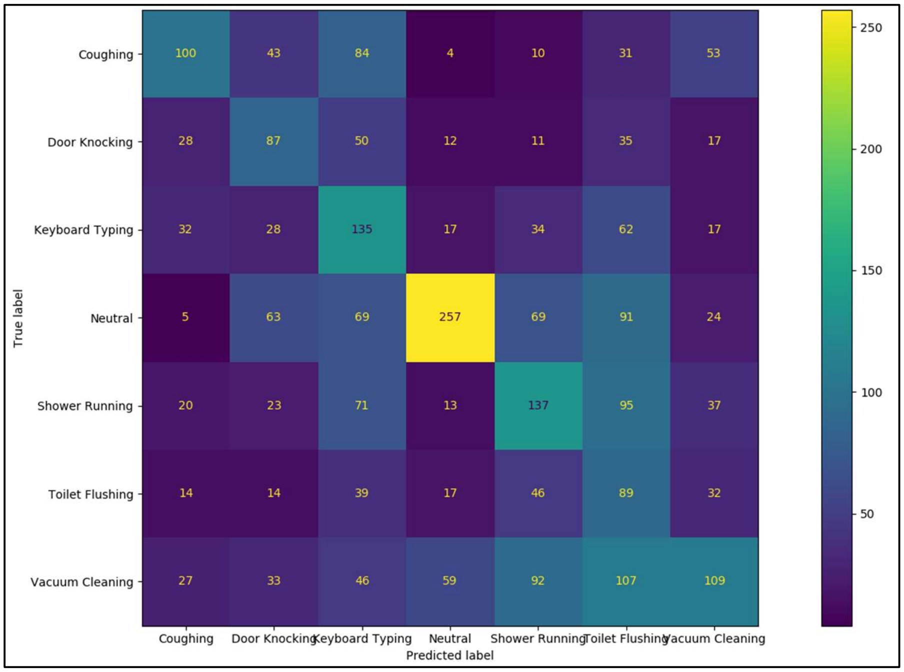

Figure III – Confusion matrix for CNN-14 model tested on validation dataset.

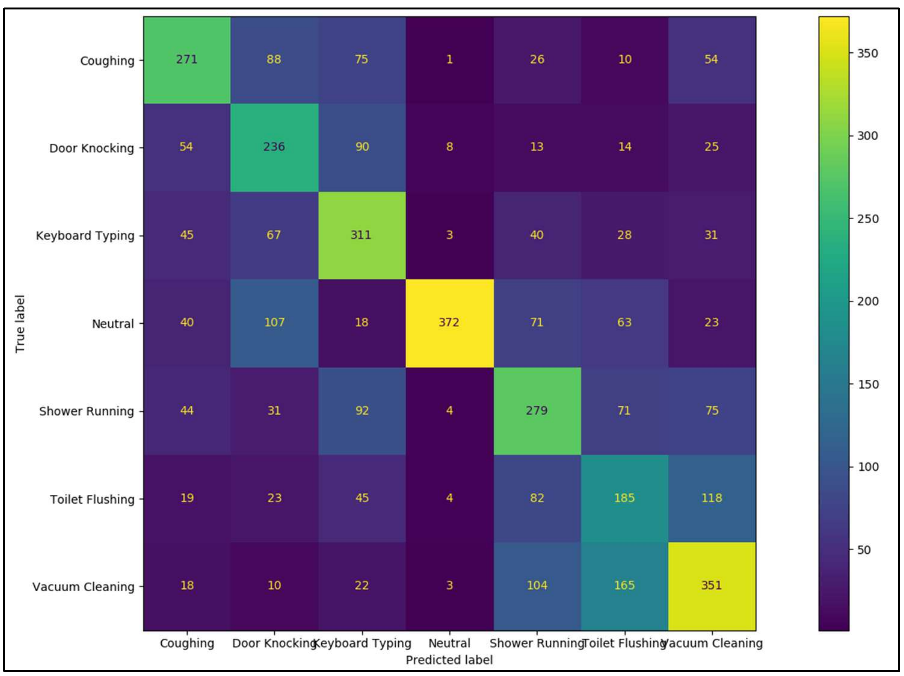

Figure IV- Confusion matrix for ResNet model tested on validation dataset

TABLE III. MICROPHONE CLASSIFICATION RESULTS

True Class	KNN Correct Prediction %	SVM Correct Prediction %	CNN-14 Correct Prediction %	ResNet Correct Prediction %	MobileNet Correct Prediction %
Coughing	40	0	0	100	0
Door Knocking	0	100	0	100	0
Keyboard Typing	0	0	100	90	0
Neutral	20	0	0	60	0
Shower Running	100	0	60	100	10
Toilet Flushing	50	0	0	80	100
Vacuum Cleaning	100	0	20	70	10

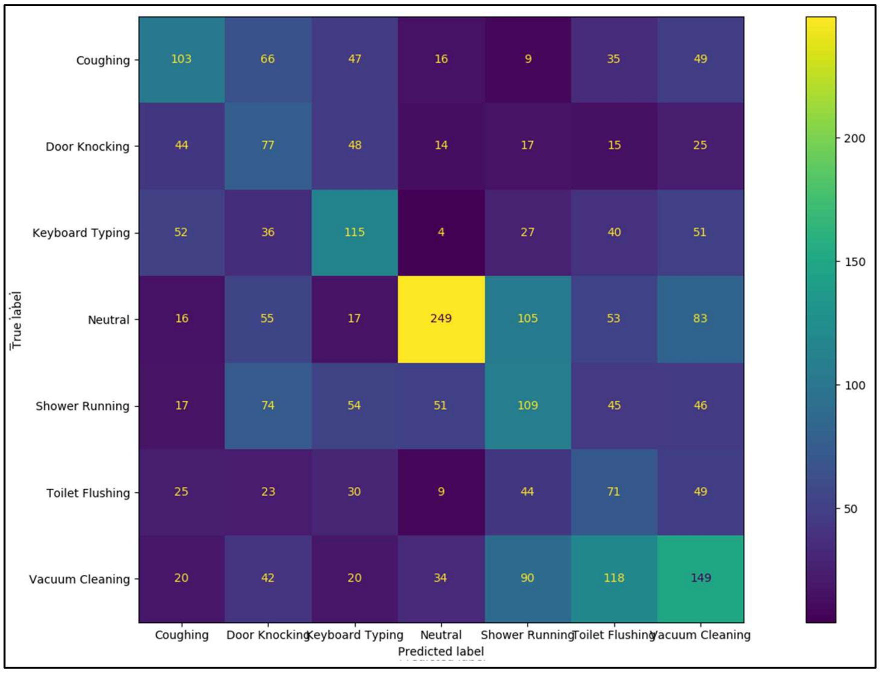

Figure V- Confusion matrix for MobileNetV2 model tested on validation dataset

input. Still, the ResNet model struggled slightly more when classifying 'continuous' sounds 'Toilet Flushing', 'Vacuum 'Keyboard Typing', 'Coughing', and also 'Shower Running'. In these classes the predictions were Cleaning' and 'Neutral'. It often got confused between these classes, and also sometimes ended up predicting one of the classes it was able to identify easier, i.e. 'coughing'.

V. Discussion

A. Class Classification

From the confusion matrixes plotted for the models (Figures I - V) it can be observed that confusion between the audio classes occurs generally in two groups. These are observed to be more 'percussive' sounds such as 'Coughing', 'Door Knocking' and 'Keyboard Typing', and then 'continuous' such as 'Neutral', 'Toilet Flushing' and 'Vacuum Cleaning'.

In general, most models seemed to be able to identify one of these types of sounds much more successfully than the other. This seems to be a result of using HPSS features as model inputs, as KNN and SVM models were investigated without these and were found to be much less successful.

The continuous sounds suffered more from class confusion, with 'Toilet Flushing' and 'Vacuum Cleaning' often being confused. This seems likely to be a result of the audio data in the dataset, as by dividing the audio samples into one second segments, only some 'Toilet Flushing' samples would have contained the actual flush moment, with the remaining samples containing audio data relating to the refilling on the basin. The water refilling sound is less distinctive than the flush sound, and much closer to white noise. With this being the case, it is much clearer why the model struggled to separate the 'Toilet Flushing' class with the 'Vacuum Cleaning' class, and also other classes with 'continuous' sounds.

The 'Neutral' class showed the most surprising behaviour as it was probably the most easily identified class in the validation data across the models, yet when the models were presented with live audio data of this class it was often misclassified. The reason for this is probably also due to the dataset audio samples. Much of the data for this class was recorded for this project using the same microphone. However, most of the data was recorded in the same room, on the

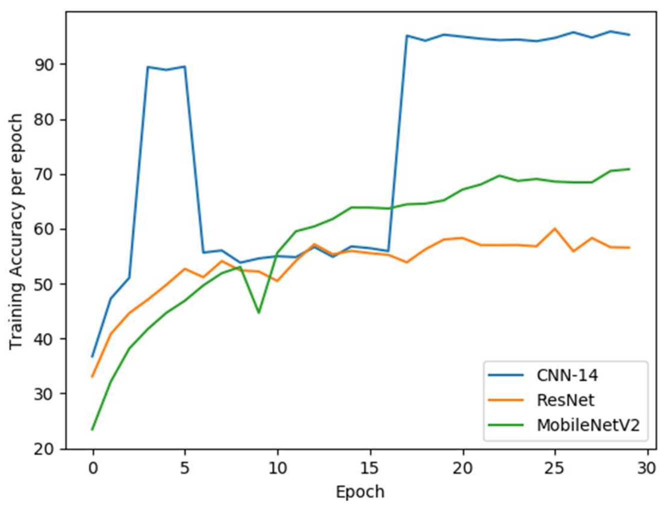

Figure VI- Training Accuracy per epoch for CNN models

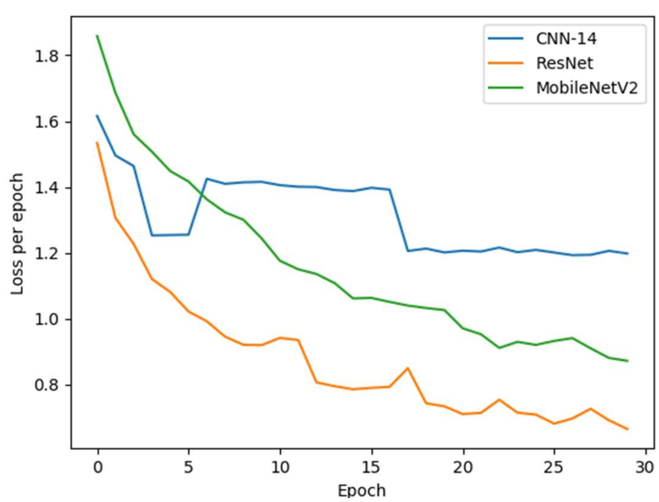

Figure VII- Training Loss per epoch for CNN models same day, data a quiet time. Perhaps this meant that much of the audio samples for this class were very similar to each other, which did not allow for variance in the class when live audio data was recorded in a different location in perhaps a noisier environment.

Interestingly, the audio class ‘Shower Running’, although seemingly a ‘continuous’ sound, was able to be classified much more accurately than the other continuous sounds. This is perhaps because ‘Shower Running’ data does contain some noticeably percussive elements, i.e. water drumming sounds as droplets hit the ground.

B. *Model size and prediction speed*

As the classification model was needed to be operational on a Raspberry Pi computer, model size and speed was a factor to consider. The non-neural network models, i.e KNN and SVM, were the smallest in size and quickest to use. The neural networks were larger due to the number of weights needed, however the fastest of these to make a prediction was the ResNet model despite it not being the smallest in size.

Whilst the Raspberry Pi 4 model was able to process all of these models without difficulty, trying to improve model efficiency may be an important consideration for future applications where a system with less processing power may be used. Also, the speed of prediction is a useful factor to consider for a real-time system, for example for assistive technology applications where a user will want to be alerted to a sound as it occurs.

C. *CNN Model Training*

From the graphs plotted during training (Figures VI and VII) it’s clear to see that the CNN model overfitted as the accuracy and loss curves suddenly jump. The MobileNet model reached a higher training accuracy than the ResNet model at around 70% accuracy, however when the model was tested on the validation data the accuracy score reached was only 33% suggesting this model also overfit.

In comparison the ResNet model showed a slow and steady climb in accuracy and a decrease in loss overtime. Still, the increase in accuracy was only very slight at each new epoch. This may have been due to data augmentation techniques. At around 20 epochs the curve seems to flatten off, probably suggesting that the model is not likely to greatly improve in accuracy past this point.

D. *Raspberry Pi System*

The difference in performance between the validation dataset and the performance of the live microphone audio input data was very noticeable. Where previously the non-CNN models had given the best accuracy, when live data was used the ResNet CNN model performed best. This seems likely to be due to the fact that the CNN models were trained using data augmentation techniques which made them more resilient to variance in class data.

The best performing models were the ResNet model and the KNN model. The KNN model seemed to perform well on the ‘continuous’ sounds, but poorly on the ‘percussive’ sounds, whilst the ResNet model showed the opposite performance. Perhaps, an ensemble combination of these two models would help to produce better accuracies.

VI. CONCLUSION

Overall, a successful system was built that was able to identify domestic sounds using a microphone and a Raspberry Pi computer. Several different models were investigated for the task, and a ResNet CNN model was found to be the most successful for use with live audio data, probably due to data augmentation techniques. The results however were not always entirely accurate, and this is likely due to training dataset issues. Percussive sound classes were most easily identified, with higher confusion in more ‘continuous noise’ classes.

To improve this system, it would be interesting to explore whether using an ensemble classification method, i.e. taking predictions from multiple models, would help improve classification accuracy, and whether this would be possible for a real-time system. Another potential area that could be further investigated is whether a different length of audio input data would lead to different classification accuracies.

Finally, this project could be extended by increasing the number of domestic sound classes for classification in order to identify more activity. A further extension could be to develop this simple system into a more useful device, for example by sending and storing this data, i.e. for remote monitoring applications, or assistive technology.

VII. REFERENCES

Abdoli, S., Cardinal, P. & Koerich, A. L., 2019. End-to-end environmental sound classification using a 1d convolutional neural network. *Expert Systems with Applications,* Volume 136, pp. 252-263.

Ahmadi, A., Dehzangi, O. & Jafari, R., 2012. Brain-computer interface signal processing algorithms: A computational cost vs. accuracy analysis for wearable computers. *2012 Ninth International Conference on Wearable and Implantable Body Sensor Networks,* pp. 40-45.

Bengio, Y., Simard, P. & Frasconi, P., 1994. Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks,* 5(2), pp. 157-166.
Boddapati, V., Petef, A., Rasmusson, J. & Lundberg, L., 2017. Classifying environmental sounds using image recognition networks. *Procedia computer science,* Volume 112, pp. 2048-2056.

Chu, S., Narayanan, S. & Kuo, C. J., 2006. *Content Analysis for Acoustic Environment Classification in Mobile Robots.* Proc. AAAI Fall Symposium.

Dai, W. et al., 2016. Very Deep Convolutional Neural Networks for Raw Waveforms. *arXiv:1610.00087.*
DCASE Challenge, 2019. *2019 Task 1 Results.* [Online] Available at: http://dcase.community/challenge2019/task-acoustic-scene-classification-results-a [Accessed June 2021].

DCASE Challenge, 2020. *2020 Task 1 Results.* [Online] Available at: http://dcase.community/challenge2020/task-acoustic-scene-classification-results-a [Accessed June 2021].

DCASE Challenge, 2021. *2021 Task 1 Results.* [Online] Available at: http://dcase.community/challenge2021/task-acoustic-scene-classification-results-a [Accessed June 2021].

E. Fonseca, X. F. J. P. F. F. &. X. S., 2020. FSD50K: An Open Dataset of Human-Labeled Sound Events.. *arXiv:2010.00475.*

Font, F., Roma, G. & Serra, X., 2013. *Freesound Technical Demo.* Barcelona, Spain, ACM International Conference on Multimedia (MM'13).

Foster, P. et al., 2015. *Chime-home: A dataset for sound source recognition in a domestic environment.* New Paltz, NY, USA , IEEE.

Gemmeke, J. F. et al., 2017. *Audio Set: An ontology and human-labeled dataset for audio events.* New Orelans, LA, USA, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).

Ghoraani, B. & Krishnan, S., 2011. Time-frequency matrix feature extraction and classification of environmental audio signals. *IEEE Trans Audio Speech Language Process,* 197(7), pp. 2197-2209.

Glorot, X. & Bengio, Y., 2010. Understanding the difficulty of training deep feedforward neural networks. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (PMLR),* Volume 9, pp. 249-256.

Goodfellow, I. et al., 2014. Generative Adversarial Networks. *Advances in Neural Information Processing Systems,* pp. 2672-2680.

Guzhov, A., Raue, F., Hees, J. & Dengel, A., 2021. *ESResNet: Environmental Sound ClassificationBased on Visual Domain Models.* Milan, Italy, 2020 25th International Conference on Pattern Recognition (ICPR).

Han, Y., Park, J. & K. Lee, 2017. Convolutional Neural Networkswith Binaural Representations and Background Subtractionfor Acoustic Scene Classification. *IEEE AASP Challenge on DCASE 2017 technical reports.*

He, K., Zhang, X., Ren, S. & Sun, J., 2016. *Deep Residual Learning for Image Recognition.* Las Vegas, NV, USA

Hershey, S. et al., 2017. CNN architectures for large-scale audio classification. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP),* pp. 131-135.

Howard, A. G. et al., 2017. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. *arXiv: 1704.04861.*

Kim, B., Chang, S., Lee, J. & Sun, D., 2021. *QTI Submission to DCASE 2021: Residual Normalization for Device-Imbalanced Acoustic Scene Classification with Efficient Design,* DCASE2021 Challenge.

Kong, Q. et al., 2020. PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing,* Volume 28, pp. 2880-2894.

Krizhevsky, A., Sutskever, I. & Hinton, G. E., 2012. ImageNet Classification with Deep Convolutional Neural Networks. *NIPS,* pp. 1106-1114.

Logan, B., 2000. *Mel frequency cepstral coefficients for music modeling.* Plymouth, USA, Proceeding of the International Symposium on Music InformationRetrieval (ISMIR).

Lu, L., Zhang, H.-J. & Li, S. Z., 2003. Content-based audio classification and segmentation by using support vector machines. *Multimedia Systems,* Volume 8, pp. 482-492.

Mahajan, S., Chatlawar, A. & Kokate, V., 2009. *Multi-class audio classification using K-nearest neighbour classifier.* National Conference on Signal and Image Processing Applications.

Mesaros, A. et al., 2018. Detection and classification of acoustic scenes and events: outcome of the DCASE 2016 challenge. *IEEE/ACM Transactions on Audio, Speech, and Language Processing,* 26(2), pp. 379-393.

Mun, S., Park, S., Han, D. K. & Ko, H., 2017. *Generative Adversarial Network Based Acoustic Scene Training Set Augmentation and Selection using SVM Hyper-Plane.*

Mun, S., Sangwook, P., David K, H. & Ko, H., 2017. Generative Adversarial Network Based Acoustic Scene Training Set Augmentation and Selection Using SVM Hyper-Plane. *Proceedings of the Detection and*

Classification of Acoustic Scenes and Events 2017 Workshop (DCASE2017), pp. 93-102.

Nguyen, T. K. T. & Pernkopf, F., 2018. Acoustic Scene Classification Using A Convolutional Neural Network Ensemble And Nearest Neighbor Filters. *Institute of Signal Processing and Speech Communication.*

Parascandolo, G., Huttunen, H. & Virtanen, T., 2016. *Recurrent Neural Networks for Polyphonic Sound Event Detection in Real Life Recordings.* Shanghai, China, International Conference on Acoustics, Speech and Signal Processing(ICASSP).

Piczak, K. J., 2015. *ESC: Dataset for Environmental Sound Classification..* Brisbane Australia, s.n., pp. 1015-1018.

Sailor, H. B., Agrawal, D. M. & Patil, H. A., 2017. Unsupervised filterbank learning using convolutional restricted boltzmann machine for environmental sound classification. *INTERSPEECH,* pp. 3107-3111.

Sakashita, Y. & Aono, M., 2018. Acoustic Scene Classification By Ensemble Of Spectrograms Based On Adaptive Temporal Division. *DCASE2018 Challenge.*

Salamon, J. & Bello, J. P., 2017. (Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification. *IEEE Signal Processing Letters,* 24(3), pp. 279-283.

Salamon, J. et al., 2017. *Scaper: A library for soundscape synthesis and augmentation.* New Paltz, NY, USA, IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA).

Sandler, M., Howard, A., M. Zhu, A. Z. & Chen, a. L.-C., 2018. MobileNetV2: Inverted Residuals and Linear Bottlenecks. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR),* pp. 4510-4520.

Sermanet, P. et al., 2014. OverFeat: Integrated Recognition, Localization and Detection using Convolutional Networks. *ICLR.*

Sharan, R. V. & Moir, T. J., 2019. Acoustic event recognition using cochleagram image and convolutional neural networks. *Applied Acoustics,* Volume 148, pp. 62-66.

Simonyan, K. & Zisserman, A., 2015. *Very Deep Convolutional Networks for Large-Scale Image Recognition.* San Diego, CA, USA

Stowell, D. et al., 2015. Detection and Classification of Acoustic Scenes and Events. *IEEE Transactions on Multimedia,* 17(10), pp. 1733-1746.

Su, Y., Zhang, K., Wang, J. & Madani, K., 2019. Environment sound classification using a two-stream CNN based on decision-level fusion. *Sensors,* 19(7), p. 1733.

Suzman, R. & Beard, J., 2015. *Global Health and Aging-Living Longer.* Bethesda, MD, USA

Szegedy, C. et al., 2015. *Going Deeper with Convolutions.* Boston, MA, USA

Tokozume, Y. & Harada, T., 2017. Learning environmental sounds with end-to-end convolutional neural network. *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP),* pp. 2721-2725.

Turpault, N., Serizel, R., Shah, A. & Salamon, J., 2019. DESED_public_eval (Version v0) [Data set]. *Zenodo. http://doi.org/10.5281/zenodo.4560759.*

United Nations, 2015. *World Population and Ageing,* New York, NY, USA: United Nations Department of Economic and Social Affairs (Population Division).

Zeiler, M. D. & Fergus, R., 2014. Visualizing and Understanding Convolutional Networks. *ECCV.*

Zhang, H., Cisse, M., Dauphin, Y. N. & Lopez-Paz, D., 2017. mixup: Beyond Empirical Risk Minimization. *arXiv:1710.09412.*

Zhang, Z., Xu, S., Cao, S. & Zhang, S., 2018. Deep Convolutional Neural Network with Mixupfor Environmental Sound Classification. *PRCV.*

Zhu, B. et al., 2018. Environmental sound classification based on multi-temporal resolution convolutional neural network combining with multi-level features. *Pacific Rim Conference on Multimedia,* pp. Springer, 528-537.